

Kwangho Baek, Ph.D.

Postdoctoral Associate in Transportation Engineering

Department of Civil, Env, and Geo- Engineering | University of Minnesota, Twin Cities | Minneapolis, Minnesota, USA | baek0040@umn.edu

RESEARCH PROFILE

Transportation engineering researcher developing AI-integrated behavioral models and data fusion frameworks for adaptive, equitable, and policy-relevant multimodal transportation planning. My work combines discrete choice modeling, machine learning, passive/active data fusion, GIS/network analysis, and transportation systems evaluation to support evidence-based decisions for transportation agencies and public-sector partners.

Research Areas	Travel Behavior; Transit Planning and Operations; AI-Integrated Discrete Choice Model; Machine Learning; Data Fusion; Mobility-as-a-Service (MaaS); Rural Transit; Equity; Shared Mobility.
Methods/Tools	Python; R; ArcGIS/QGIS; SQL; EMME4; TransCAD; Synchro; Machine Learning Frameworks (PyTorch, Scikit-Learn); Discrete Choice Modeling; Spatial and Network Analysis.

EDUCATION

Sep 2021–Feb 2026	Ph.D. in Transportation Engineering - Dissertation: <i>Toward Adaptive Transit Planning with Artificial Intelligence</i> . - Advisor: Dr. Alireza Khani. GPA: 3.905/4.000. - Master of Science in Civil Engineering conferred from the same program in May 2023.	University of Minnesota, Twin Cities
Mar 2018–Feb 2020	Master of Science in Urban Planning and Engineering - Thesis: <i>The Impact of the Bus Crowding Information System on Passenger Bus Choice</i> . - Advisor: Dr. Jin-Hyuk Chung. GPA: 4.21/4.30.	Yonsei University, Seoul
Mar 2012–Feb 2018	Bachelor of Science in Engineering - Capstone: <i>A Study on the Urban Development in North Korea After Reunification: Focused on Micro-districts in Pyongyang</i> . Co-authors: Hyunmin Daniel Zoh, Subin Park, and Jihyeon Ock. - GPA: 3.76/4.30, <i>magna cum laude</i> .	Yonsei University, Seoul

PEER-REVIEWED PUBLICATIONS

Kwangho's name is underlined; *: Corresponding author

Published or In Press

- **Navigating Machine Learning for Travel Behavior Analysis: A Comprehensive Guide to Inferring Trip Purposes Using Transit Survey and Automatic Fare Collection Data Fusion**
- Kwangho Baek and Alireza Khani*
- *Transportation Research Part C: Emerging Technologies*. Vol 190 (2026), 105726.
- **Evaluating the Effects of Mobility-as-a-Service Platform in Increasing Equity and Transit Ridership in Rural Areas**
- Kwangho Baek and Alireza Khani*
- *Transportation Research Record*. Vol Online First (2026).
- Also presented at the Transportation Research Board 104th Annual Meeting. Washington, DC. Jan 6, 2025.
- **Charger Deployment and Fleet Sizing for Heterogeneous Fleets Under Budget-Constrained Multi-Stage Electrification**
- Nastaran Tork, Behnam Emami, Kwangho Baek, Alireza Khani*, Sushmita Bhandari, and Alyssa Ryan
- *Transport Policy*. Vol 187 (2026), 104285.
- Also presented at the Transportation Research Board 105th Annual Meeting. Washington, DC. Jan 12, 2026.
- **Efficient Urban Transit Systems: Bridging Freight Logistics and Passenger Transport**
- Nastaran Tork, Kwangho Baek, and Alireza Khani*
- *Transportation Research Record*. Vol 2679(11) (2025), 782–804.
- Also presented at the Transportation Research Board 104th Annual Meeting. Washington, DC. Jan 6, 2025.
- **Electric Scooter Sharing: How Do People Value It as a Last-Mile Transportation Mode?**
- Kwangho Baek, Hyukseong Lee, Jin-Hyuk Chung, and Jinhee Kim*
- *Transportation Research Part D: Transport and Environment*. Vol 90 (2021), 102641.
- **Factors Affecting Heterogeneity in Willingness to Use E-Scooter Sharing Services**
- Hyukseong Lee, Kwangho Baek, Jin-Hyuk Chung, and Jinhee Kim*
- *Transportation Research Part D: Transport and Environment*. Vol 92 (2021), 102751.
- Also presented at the 81st Korean Society of Transportation Conference. Yeosu, South Korea. Sep 27, 2019.

Under Review or Revision

- **Supporting High-Speed Rail with a Hub-and-Shuttle Approach: Transforming Long-Haul Conventional Rails into Short and Frequent Feeders**
 - Ju-Yeon Lee, Kyungtaek Kim, and Kwangho Baek*
 - Under Revision, *Transportation Research Procedia*.
 - Also presented at the World Conference on Transport Research (WCTR). Toulouse, France. Jul 10, 2026.
- **A Discrete Choice Model with Segmentation and Embedding-Augmented Learning for Travel Behavior Analysis**
 - Kwangho Baek and Alireza Khani*
 - Under Review, *Transportation*.
 - Also presented at the Conference on Advanced Systems in Public Transport (CASPT). Kyoto, Japan. Jul 1, 2025.

CONFERENCE PRESENTATIONS (PEER-REVIEWED)

- **Unlocking Vehicle-Miles-Traveled Potential: Behavioral Segmentation and Trip-Context Effects from a Machine Learning-Augmented Mode Choice Model**
 - Kwangho Baek and Alireza Khani*
 - (Accepted) Transportation Research Board 106th Annual Meeting. Washington, DC. Jan 2027.
- **Learning-Based Intelligent and Adaptive Dispatching for MaaS-Enabled Intermodal Rural Transit**
 - Kwangho Baek and Alireza Khani*
 - (Accepted) Transportation Research Board 106th Annual Meeting. Washington, DC. Jan 2027.
- **Evaluating Incremental Truck Weight Limit Policy at the Statewide Network Scale: A Minnesota Case Study**
 - Nastaran Tork, Alireza Khani*, Kwangho Baek, Mihai Marasteanu
 - (Accepted) Transportation Research Board 106th Annual Meeting. Washington, DC. Jan 2027.
- **Quantifying Pavement Damage Costs Associated with Increased Axle Loads Using Life-Cycle Cost Analysis and MnPAVE Software**
 - Zifeng Zhao*, Kaidi Liang, Kwangho Baek, Mihai Marasteanu, and Alireza Khani
 - (Accepted) Transportation Research Board 106th Annual Meeting. Washington, DC. Jan 2027.
- **Enhancing Rural Transit Efficiency through MaaS Integration of Demand-Responsive Transit and Fixed Routes**
 - Kwangho Baek and Alireza Khani*
 - ASCE International Conference on Transportation and Development (ICTD). Detroit, MI. Jul 1, 2026.
- **Dynamic Evolution of MaaS Subscriptions: A Game-Theoretic Day-to-Day Framework for Multimodal Transport**
 - Jiale Li, Alireza Khani*, and Kwangho Baek
 - ASCE International Conference on Transportation and Development (ICTD). Detroit, MI. Jun 30, 2026.
- **Barriers to Local Government Electric Vehicle Fleet Transitions: Insights from Minnesota**
 - Sushmita Bhandari, Juliana Panhorst, Alyssa Ryan*, Nastaran Tork, Kwangho Baek, and Alireza Khani
 - Transportation Research Board 105th Annual Meeting. Washington, DC. Jan 12, 2026.
- **The Changes in Transit Route Choice Behavior Between Pre- and Post-Pandemic**
 - Sahas Sok, Kwangho Baek, and Alireza Khani*
 - Transportation Research Board 104th Annual Meeting. Washington, DC. Jan 8, 2025.
- **Integrating Neural Networks and Latent Class Choice Model to Analyze Transit Path Choice: Understanding Likely Users of Branded or Premium Transit Routes**
 - Kwangho Baek and Alireza Khani*
 - Transportation Research Board 104th Annual Meeting. Washington, DC. Jan 7, 2025.
- **Changes in Pre- and Post-Pandemic Transit Commute Trips: Findings from Integrating Smart Card Data and Passenger Surveys with Machine Learning Models**
 - Kwangho Baek and Alireza Khani*
 - Transit Data 2024. London, United Kingdom. Jul 3, 2024.
- **A DRT-Bus Intermodal Rural Transit Operation Powered by MaaS Applying Differentiated Dispatch Priorities**
 - Kwangho Baek and Alireza Khani*
 - Transportation Research Board 103rd Annual Meeting. Washington, DC. Jan 9, 2024.
- **How Do People Value Reliable Modes in Their Transit Path Choices?**
 - Kwangho Baek and Alireza Khani*
 - International Symposium on Transport Network Resilience (INSTR). Hong Kong. Dec 13, 2023.
- **How Has Real-Time Crowding Information Affected Passengers' Bus Choice?**
 - Kwangho Baek and Jin-Hyuk Chung*
 - Transportation Research Board 101st Annual Meeting. Washington, DC. Jan 12, 2022.
- **A Practical Method to Adjust Bus Routes Based on Transfer Penalties Using Trip-Chain Data and SP Survey**
 - Younghun Bahk, Kwangho Baek, and Jin-Hyuk Chung*
 - Transportation Research Board 100th Annual Meeting. Washington, DC (virtual). Jan 28, 2021.

INVITED TALKS

- **Longitudinal Analysis of Transit Rider Behavior Using AFC and Survey Data Fusion via Machine Learning: COVID-19-Driven Commute Pattern Changes**
 - Kwangho Baek and Alireza Khani
 - Transportation Research Board 105th Annual Meeting. Washington, DC. Jan 12, 2026.
- **Impacts of Expanding Transit Data Access for Rural Transit in Minnesota**
 - Elliott McFadden, Kwangho Baek, Hannah DeBruin, and Alireza Khani
 - 2024 CTS Transportation Research Conference. Minneapolis, MN. Nov 7, 2024.

TECHNICAL REPORTS

- **Transitioning to EV Fleets: Best Practices and a Decision Tool**
 - Nastaran Tork, Kwangho Baek, Alireza Khani, Sushmita Bhandari, and Alyssa Ryan
 - *Minnesota Department of Transportation* (2026). Report No. MN 2026-15.
- **Transportation Needs and Current Services in the 5-County East Central Region**
 - Kwangho Baek, Theodore Jessup, Frank Douma, Alireza Khani, and Michael Darger
 - *Center for Transportation Studies, University of Minnesota* (2025). Report No. ESMC 26-01.
- **Transfer Behavior and Off-Peak Commutes**
 - Kwangho Baek and Alireza Khani
 - *Center for Transportation Studies, University of Minnesota* (2024). Report No. CTS 24-06.
- **MnDOT's Mobility-as-a-Service Platform: Assessing User Behavior and Measuring System Benefits**
 - Kwangho Baek, Hannah DeBruin, Alireza Khani, and Elliott McFadden
 - *Center for Transportation Studies, University of Minnesota* (2024). Report No. CTS 24-01.

ACADEMIC APPOINTMENTS AND RESEARCH EXPERIENCE

Mar 2026–Present	Postdoctoral Associate, Transit Lab • Managing research projects conducted by graduate students and interns in the lab. • Developing research proposals and pursue funding opportunities. • Developing end-to-end multimodal routing, behavioral modeling, and generative AI frameworks to support evidence-based planning and policy decisions. Funded by Minnesota DOT. • Investigating the effects of transportation infrastructure on economic development with benefit-cost analysis, accessibility modeling, and simulation. Funded by Minnesota DOT.	University of Minnesota, Twin Cities
Aug 2021–Feb 2026	Graduate Assistant and Lab Manager, Transit Lab • Collaborated on research projects led by graduate students and interns; managed lab equipment, servers, databases, and software tools. • Led a regional transit needs assessment with spatial gap analysis in East Central Minnesota. Funded by the Empowering Small Minnesota Communities Program. • Assessed rural Minnesota Mobility-as-a-Service deployment outcomes. Funded by Minnesota DOT and the Federal Transit Administration. • Investigated enhanced transit services' impacts on off-peak commutes by developing transit path-finding algorithms and enriching automatic fare collection data with machine learning models. Funded by the Metropolitan Council, Minnesota.	University of Minnesota, Twin Cities
Dec 2019–Jul 2021	Researcher, Department of Rail Transport • Conducted demand and feasibility analysis for The Fourth National Railway Plans of Korea (2021–2030) using EMME4. • Proposed operation-speed-based resource reallocation for Korean National Rail, optimizing a mixed low- and high-speed rail network under vehicle and budget constraints.	Korea Transport Institute (KOTI)
Feb 2018–Dec 2019	Graduate Assistant, Transportation Planning Lab • Constructed a structural equation model to analyze factors influencing transit user satisfaction nationwide. • Conducted network analyses to identify an optimal national firefighters' hospital location, considering weighted multimodal access time, revenue, and healthcare equity.	Yonsei University, Seoul
Aug 2017–Feb 2018	Research Assistant • Engineered an ArcGIS geoprocessing algorithm for identifying geospatial clusters of large-scale social media location data. • Designed automation criteria for identifying borders and catchment areas of social media tag-based commercial zones.	Barun ICT Research Center

Dec 2016–Mar 2017	Undergraduate Intern, Environmental Spatial Informatics Lab - Developed ArcGIS lecture materials for a regular undergraduate 3000-level course. - Supported NLP-based text mining to extract disaster events from historical news data.	Yonsei University, Seoul
Apr 2014–Jan 2016	Terrain and GIS Analyst, Military Engineer Sergeant - Managed GIS databases and created on-demand tactical maps for an infantry division. - Developed an ArcGIS algorithm to detect eligible battlefield water points.	Republic of Korea Army

TEACHING EXPERIENCE

Fall 2022; Fall 2025	Lab Instructor, CEGE 3201: Transportation Engineering - Taught weekly one-hour lab sessions in English, including R and Synchro labs for traffic engineering and transit data analysis. - Average teaching evaluation across two semesters: 5.48/6.00. - Feedback (Fall 2025): <i>“He breaks down concepts clearly and takes time to work through examples step by step. His patience with students who have no coding background really stands out.”</i> - Feedback (Fall 2022): <i>“I liked that he would guide us through how to use commands in R. He explained everything he was doing and made it easy to follow along. Great instructor!”</i>	University of Minnesota, Twin Cities
Spring 2019	Lab Instructor, CRP 3000: Transportation System Analysis Taught weekly one-hour lab sessions in Korean on R and TransCAD applications for four-step transportation planning.	Yonsei University, Seoul
Spring/Fall 2018	Teaching Assistant - CRP 2320: Transportation Planning (undergraduate) - CRP 6561: Transportation Planning (graduate) - CRP 3660: Site Transportation Planning	Yonsei University, Seoul

MISCELLANEOUS

Journal and Conference Reviewing

- Transportation Research Part D: Transport and Environment:** Reviewed 2 full journal manuscripts.
- Transportation:** Reviewed 1 full journal manuscript.
- Transportation Letters:** Reviewed 1 full journal manuscript.
- Transportation Research Today:** Reviewed 1 full journal manuscript.
- Transportation Research Board (TRB) Annual Meeting:** Reviewed 12 full conference papers.
- International Symposium on Transport Network Resilience:** Reviewed 2 extended abstracts.

Academic Service and Professional Membership

Jun 2026–Dec 2029	World Conference on Transport Research Society (WCTRS)	Member
Feb 2026–Dec 2026	American Society of Civil Engineers (ASCE)	Student Member→Member
Jan 2022–Jan 2026	Korean-American Scientists and Engineers Association (KSEA)	Student Member
Sep 2024–Aug 2025	UMN CEGE Graduate Student Council	Academic Chair/Transportation Representative
Mar 2018–Dec 2019	Korean Society of Transportation (KST)	Student Member

Honors and Awards

2026	Robert C. Johns Research Partnership Award. (team award) Center for Transportation Studies.
2025	Matthew J. Huber Award for Excellence in Transportation Research. Center for Transportation Studies.
2025; 2022	Best Presentation Awards. CEGE 8200 Transportation Seminar, University of Minnesota
2024	Excellent Presentation Award. Korea Transportation Association in America.
2019	Brain Korea 21c Fellowship. Korea Ministry of Education.
2018	President’s Award. Korean Society of Transportation.